POWER AUTOMATION TEAM

DCS & INSTRUMENTS SPECIALISTS

COMPANY PRESENTATION · 2026

SUCCESS STORIES

28 reference projects in Egypt, Saudi Arabia and Oman — DCS commissioning, loop checks, start-up and training since 1991.

patdcs.com · +20 109 991 2020 · info@patdcs.com

TRUSTED BY

CONTRACTORS, VENDORS AND PLANT OWNERS

Companies PAT engineers and technicians have delivered for since 1991.

EPC CONTRACTORS & SYSTEM VENDORS

EMERSON · Hanwha · DAEWOO E&C · GE · ORASCOM CONSTRUCTION · Iberdrola · KBR

John Cockerill · DOOSAN · TECNICAS REUNIDAS · GIZA SYSTEMS · SAUDI FAL · SAVOLA · L3ORON Institute

PLANT OWNERS & END USERS

أرامكو السعودية saudi aramco · الشركة السعودية للكهرباء Saudi Electricity Company نعمل بإتقان من أجلكم · مرافق MARAFIQ · وزارة الكهرباء والطاقة المتجددة · PHOENIX POWER · وزارة الكهرباء والطاقة المتجددة هيئة الطاقة الجديدة والمتجددة · UNITED SUGAR

PAT IN NUMBERS

THE COMMISSIONING SPECIALISTS

1991	Established in Egypt
28	Reference projects
3	Countries: Egypt, Saudi Arabia, Oman
12 GW+	Combined capacity of the plants we worked on
24/7	Support for our clients



SECTORS WE SERVE

INDUSTRIES WE COMMISSION

8

STEAM POWER PLANTS

Ataka Power Plant 2×300 + 2×150 MW, Shoaiba ST Power Plant 3×250 MW, Rabigh Power Plant 4×300 MW

8

COMBINED-CYCLE POWER PLANTS

Damietta CCPP 1,230 MW, New Talkha & El Kurimat II CCPP, New Talkha CCPP 750 MW

5

GAS-TURBINE POWER PLANTS

PP4 Riyadh Power Plant, SEC4A Al Jouf, SEC4A Al Quryat

3

WATER & UTILITIES

Shoaiba lime water plant, Marafiq Yanbu II 3×270 MW, Yanbu I & II

1

OIL & GAS PROCESSING

Abqaiq plants

1

SOLAR (ISCC)

Solar Energy Kurimat Power Plant

2

INDUSTRIAL & MARINE

Suez Sokhna Port United Sugar Project, El Esraa petroleum tanker ship, Alexandria

Counted by reference project.

Most of our work is DCS commissioning on power plants — and the same procedures carry over to water, oil & gas and industrial sites.



WHERE WE HAVE WORKED

THREE COUNTRIES, ONE TEAM

14 EGYPT
reference projects

13 SAUDI ARABIA
reference projects

1 OMAN
reference projects

Pins show plant locations; numbers are projects per site.

2004–2020

EGYPT

14 REFERENCE PROJECTS

Damietta · Talkha · El Kureimat · Ataka (Suez) · Ain Sokhna · Al Arish · Alexandria

Steam and combined-cycle power plants, solar ISCC, industrial and marine.



EGYPT · 2004 · EMERSON OVATION

ATAKA POWER PLANT 2×300 + 2×150 MW

DCS retrofit

CONTRACTOR

EMERSON

Emerson

END USER

Egyptian Electricity Holding Co.

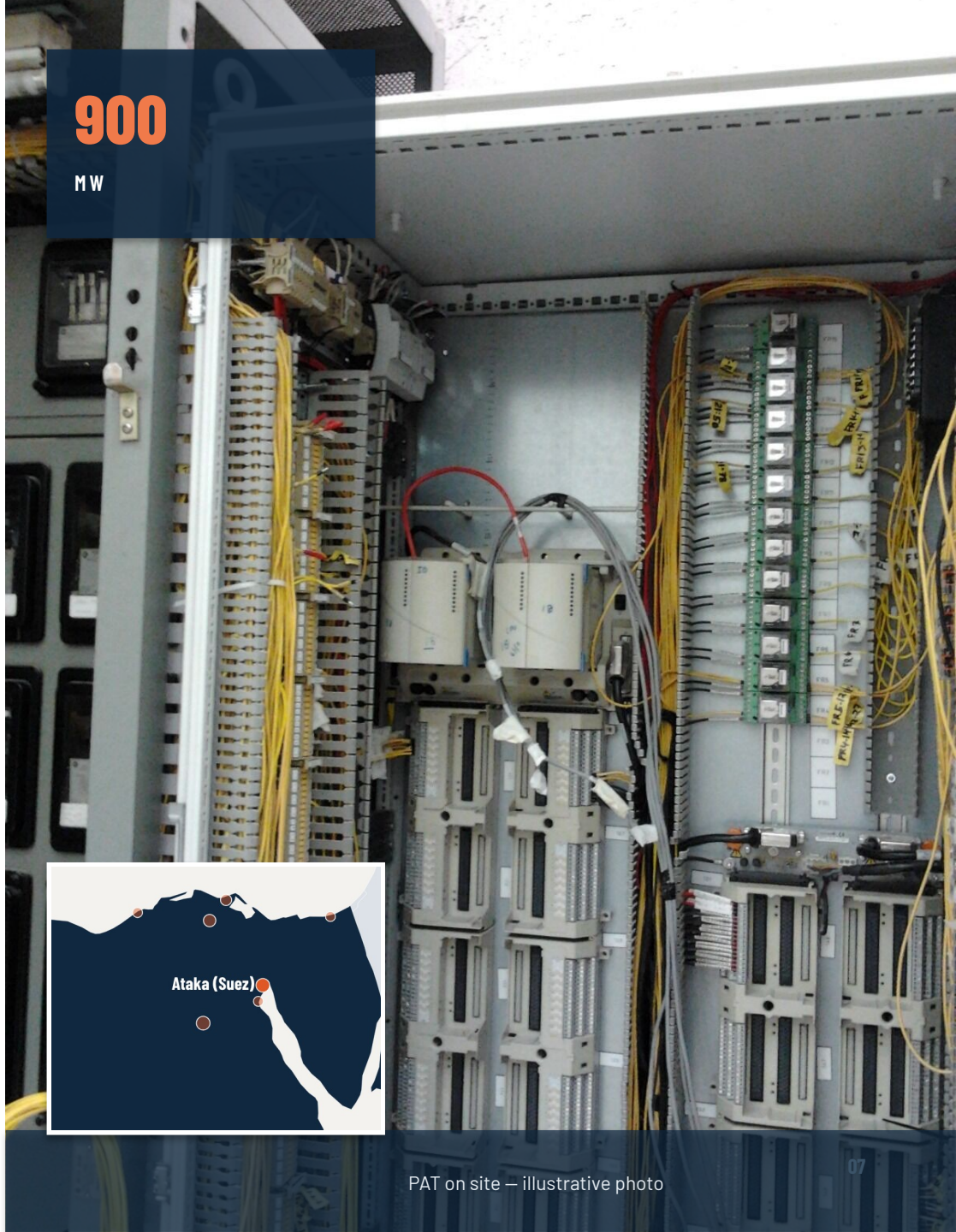
THE PROJECT

DCS retrofit of a steam power plant: old panels removed, new DCS installed and units returned to commercial operation.

PAT SCOPE

- Removal of the old panels and supervision of the new DCS installation and replacement
- Commissioning and start-up, returning the DCS units to commercial operation
- Owner staff training courses on DCS plant operation and maintenance

RESULT DCS units returned to commercial operation; owner staff trained.



PAT on site — illustrative photo

EGYPT · 2006 · EMERSON OVATION

DAMIETTA CCPP 1,230 MW

DCS migration

CONTRACTOR

Emerson

END USER

Egyptian Electricity Holding Co.

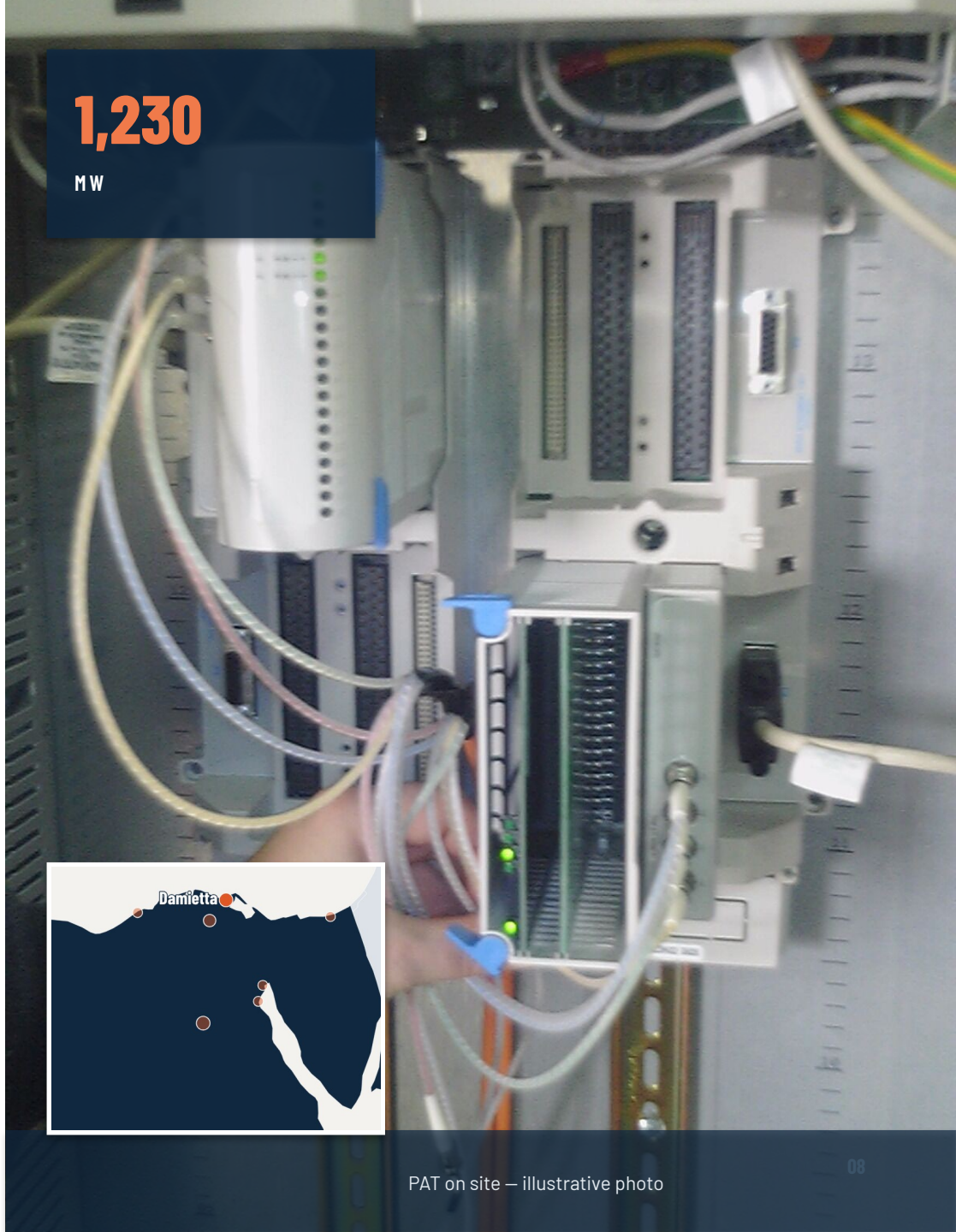
THE PROJECT

Migration of the plant DCS from Westinghouse WDPF to Emerson Ovation, with commissioning, start-up and owner training.

PAT SCOPE

- Installation and commissioning of the new DCS parts
- Commissioning and start-up, returning the DCS units to commercial operation
- Owner staff training courses on DCS plant operation and maintenance

RESULT Units returned to commercial operation on Ovation; owner staff trained.



PAT on site — illustrative photo

EGYPT · 2007–2009 · EMERSON OVATION

NEW TALKHA & EL KURIMAT II CCPP — 2 × 750 MW

DCS installation & commissioning

CONTRACTOR

Emerson · SKODA Power (now Doosan Skoda Power) · CMI (now John Cockerill)

END USER

Egyptian Electricity Holding Co.

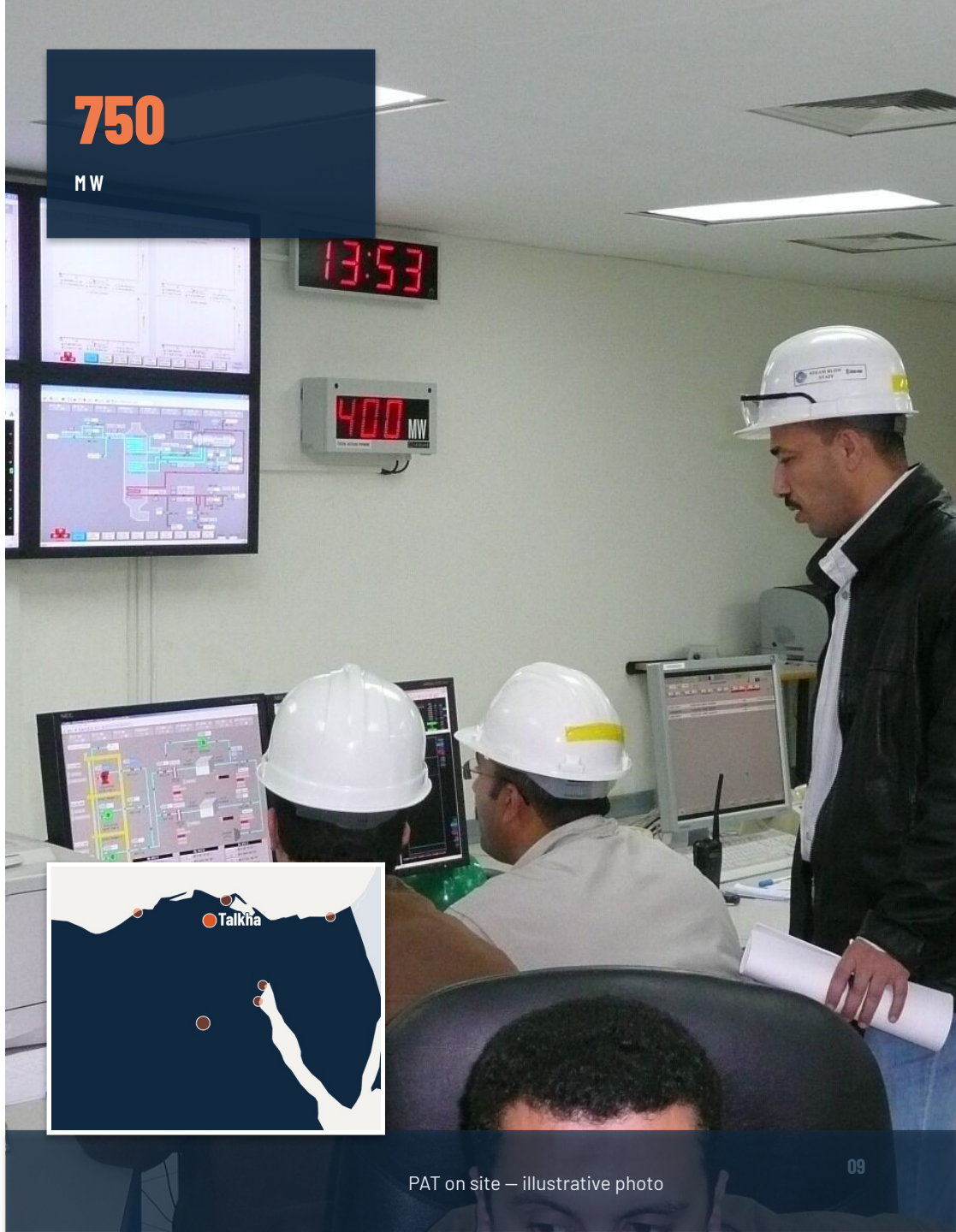
THE PROJECT

DCS installation, hardware and software modifications and configuration for normal operation on two 750 MW combined-cycle plants.

PAT SCOPE

- Emerson — DCS installation & start-up
- SKODA Power — BOP commissioning
- CMI — HRSG commissioning

RESULT Both plants powered up and configured for normal operation; owner staff trained.



PAT on site — illustrative photo

EGYPT · 2009 · SIEMENS S7 PLC

SUEZ SOKHNA PORT UNITED SUGAR PROJECT

PLC commissioning

CONTRACTOR

Savola Group

END USER

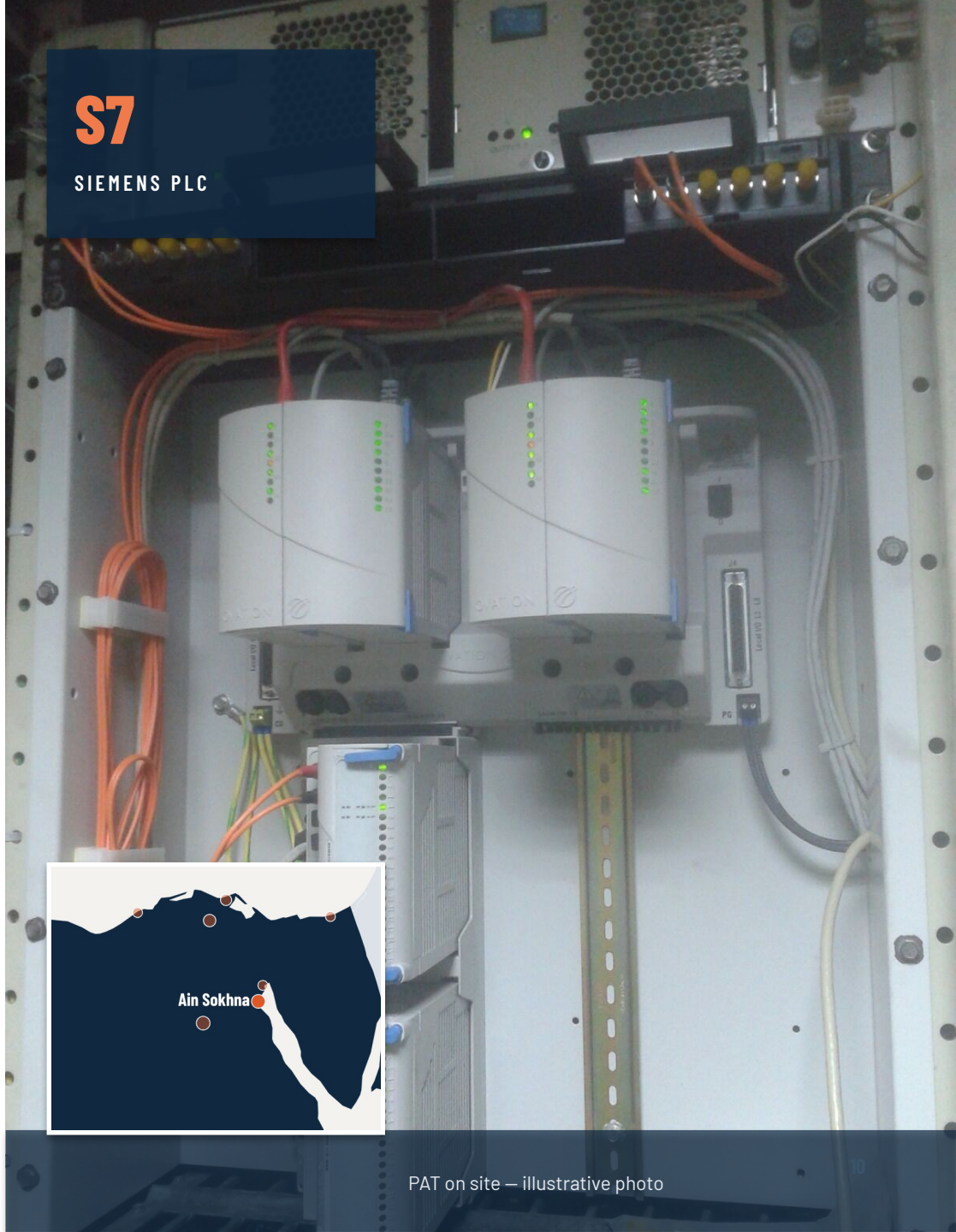
United Sugar Company of Egypt

THE PROJECT

IO loop and device checks and commissioning of a Siemens S7 PLC system for Savola (Saudi Arabia).

PAT SCOPE

- Siemens S7 PLC IO loop and device checks and commissioning



PAT on site — illustrative photo

EGYPT · 2009

NEW TALKHA CCPP 750 MW — CWP

CWP commissioning & start-up

CONTRACTOR

TECNICAS REUNIDAS

INITEC Energia (Tecnicas Reunidas group)

END USER

Egyptian Electricity Holding Co.

THE PROJECT

CWP commissioning, start-up and operation scope for INITEC (Spain) at the New Talkha combined-cycle plant.

PAT SCOPE

- CWP commissioning, start-up and operation scope



PAT on site — illustrative photo

EGYPT · 2010 · ALSTOM DCS

EL KURIMAT III 750 MW

I&C commissioning

CONTRACTOR

TECNICAS REUNIDAS

INITEC Energia (Tecnicas Reunidas group)

END USER

Egyptian Electricity Holding Co.

THE PROJECT

IO loop and I&C device commissioning team for the Alstom DCS, working for INITEC (Spain).

PAT SCOPE

- Alstom DCS IO loop and I&C device commissioning team



PAT on site — illustrative photo

EGYPT · 2011–2012 · EMERSON DCS

AL ARISH STPP 2×30 MW

DCS migration

CONTRACTOR

Emerson

END USER

Egyptian Electricity Holding Co.

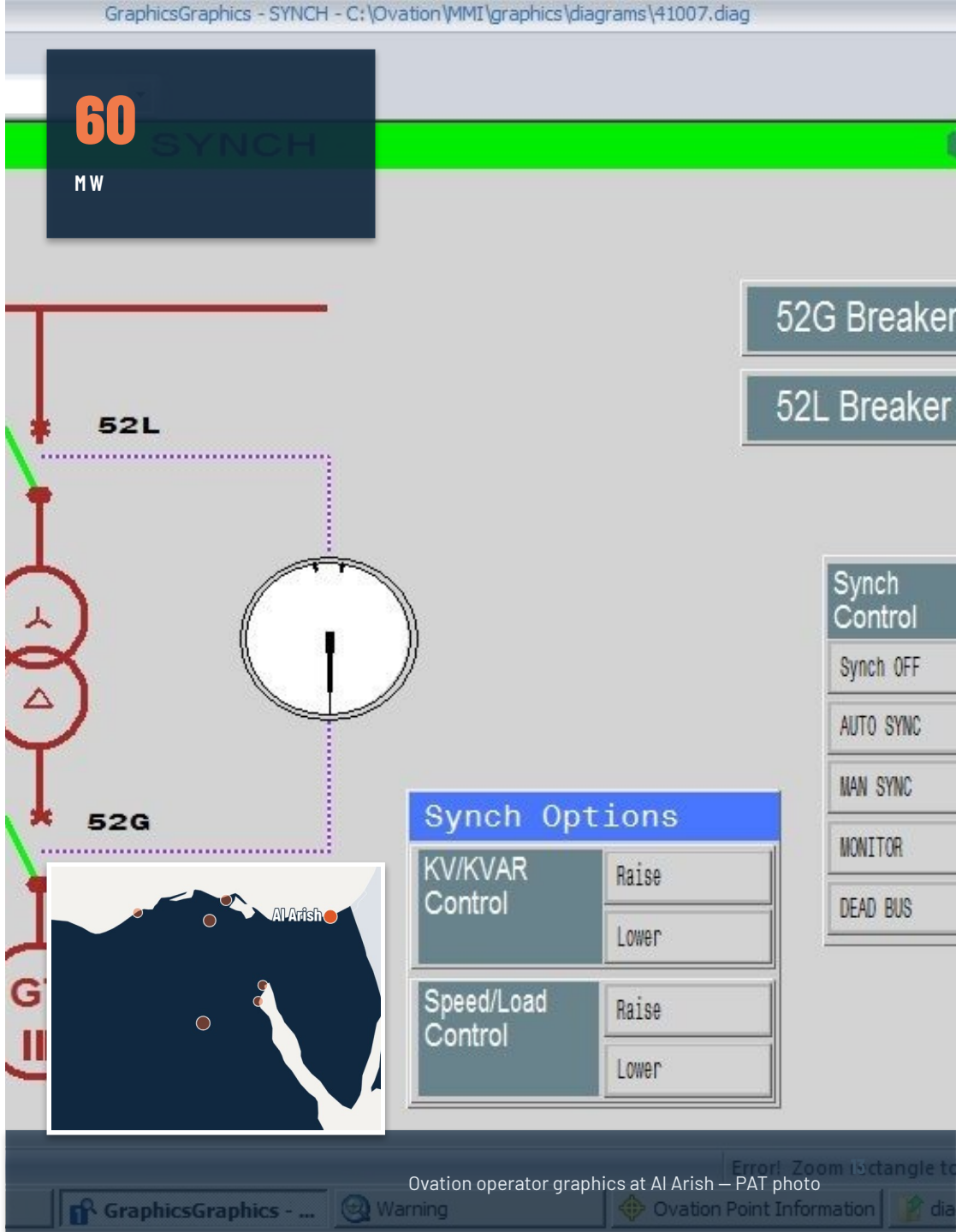
THE PROJECT

Complete DCS removal and installation scope and migration activities on a steam power plant.

PAT SCOPE

- Complete DCS removal and installation scope and migration activities
- Plant device check and test, commissioning of plant systems and return to normal operation
- Owner staff training courses on DCS plant operation and maintenance

RESULT Plant systems back to normal operation; owner staff trained.



Ovation operator graphics at Al Arish — PAT photo

EGYPT · 2011 · EMERSON OVATION

OLD TALKHA 210 MW — TCS UNIT 2

DCS retrofit

CONTRACTOR

EMERSON

Emerson

END USER

Egyptian Electricity Holding Co.

THE PROJECT

Removal of the old TCS control cabinets and complete installation of a new Ovation DCS on Unit 2.

PAT SCOPE

- Removal of the old TCS control cabinets and complete installation scope of the new Ovation DCS

RESULT New Ovation DCS installed on Unit 2.



PAT on site — illustrative photo

EGYPT · 2011

SOLAR ENERGY KURIMAT POWER PLANT

Plant commissioning & start-up

CONTRACTOR

Iberdrola

END USER

وزارة الكهرباء والطاقة المتجددة
هيئة الطاقة الجديدة والمتجددة

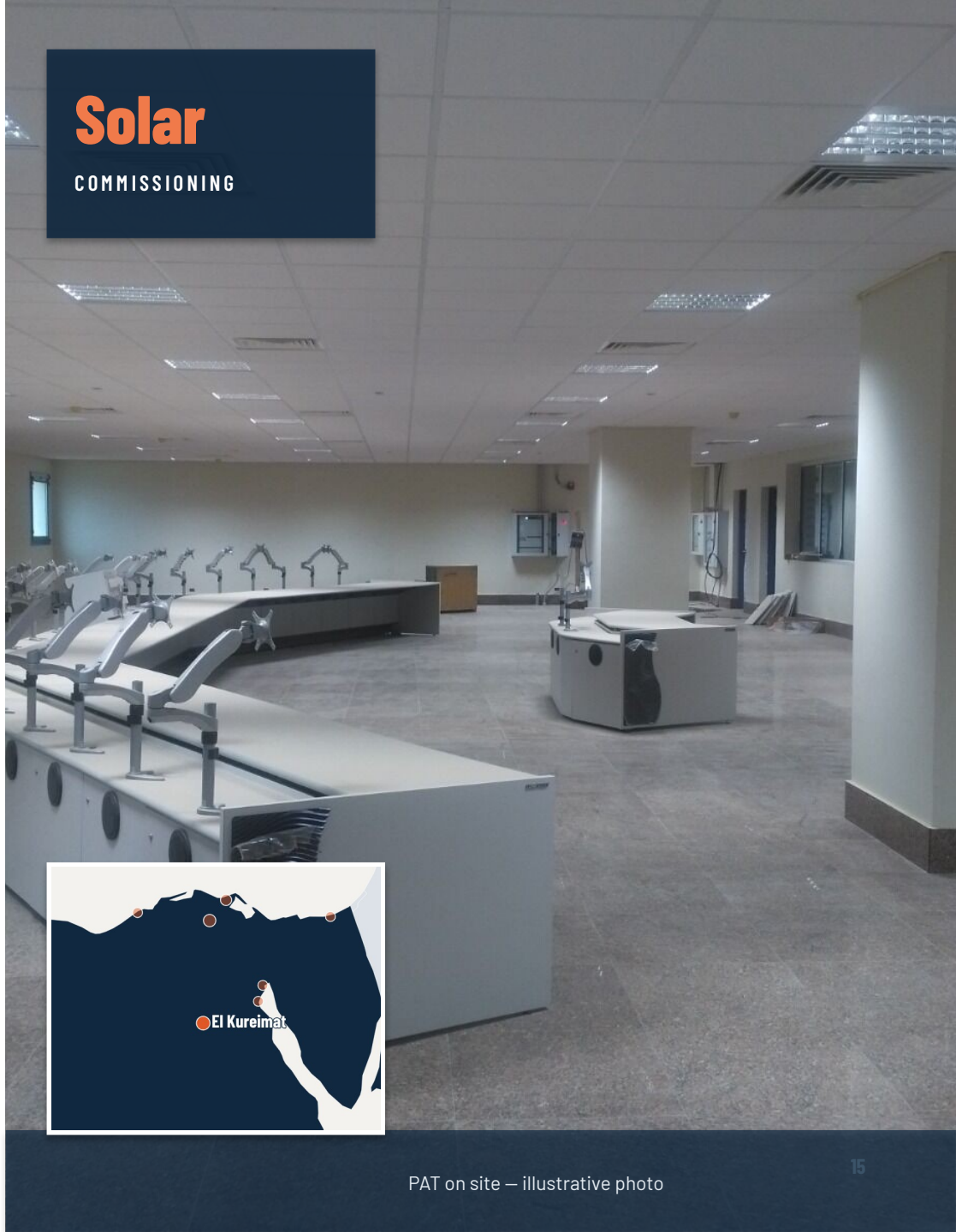
NREA — New & Renewable Energy Authority

THE PROJECT

Plant commissioning, start-up and operation within Iberdrola's scope.

PAT SCOPE

- Plant commissioning, start-up and operation within the Iberdrola scope



PAT on site — illustrative photo

EGYPT · 2015–2016 · EMERSON DCS

EL KURIMAT I 2×650 MW

DCS migration

CONTRACTOR

Emerson

END USER

Egyptian Electricity Holding Co.

THE PROJECT

Site management and supervision of the DCS replacement on a very tight schedule, through start-up and return to commercial operation.

PAT SCOPE

- Site management and supervision, inspecting and checking out the removal of the old DCS parts and the installation of the new DCS, on a very tight schedule
- Start-up and commissioning of the new DCS, powering up and configuring it for normal operation
- Plant device checks and tests (database, graphics, logic) and hardware and software modifications

RESULT DCS back in normal commercial operation on a very tight schedule.



PAT on site — illustrative photo

EGYPT · 2017

WEST DAMIETTA POWER STATION CCPP — GT 4×125 + 1×250 MW

I&C commissioning

CONTRACTOR

GE · Orascom Construction

END USER

Egyptian Electricity Holding Co.

THE PROJECT

I&C commissioning scope for four HRSGs, BOP, STG and ACC plants: cold and hot IO loop tests and plant device functional tests.

PAT SCOPE

- I&C commissioning scope for 4 HRSG, BOP, STG and ACC plants
- IO cold and hot loop tests
- Plant device functional tests



PAT on site — illustrative photo

EGYPT · 2020

EL ESRAA PETROLEUM TANKER SHIP, ALEXANDRIA

Electrical & PLC troubleshooting

CONTRACTOR

DIRECT CLIENT

Direct contract

END USER

EL ESRAA

El Esraa Co.

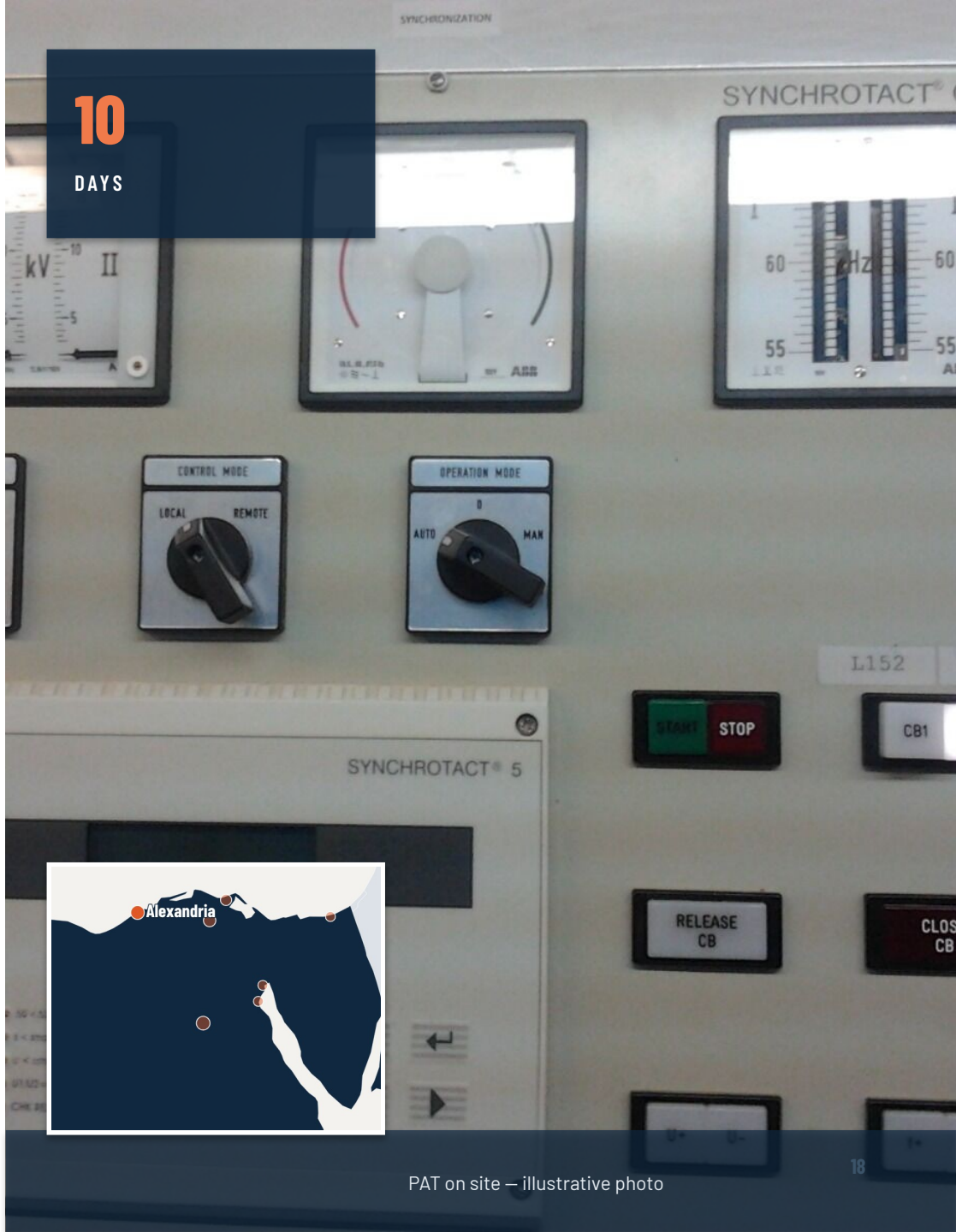
THE PROJECT

Troubleshooting of generator systems, switchgear, emergency systems and PLCs; the ship was handed over to the first engineer within 10 days after standing idle for years.

PAT SCOPE

- Troubleshooting of the electrical generator systems, switchgear, emergency systems and PLCs
- Task completed and the ship handed over to the first engineer within 10 days, after being stopped for years

RESULT Ship handed over to the first engineer within 10 days after standing idle for years.



PAT on site — illustrative photo

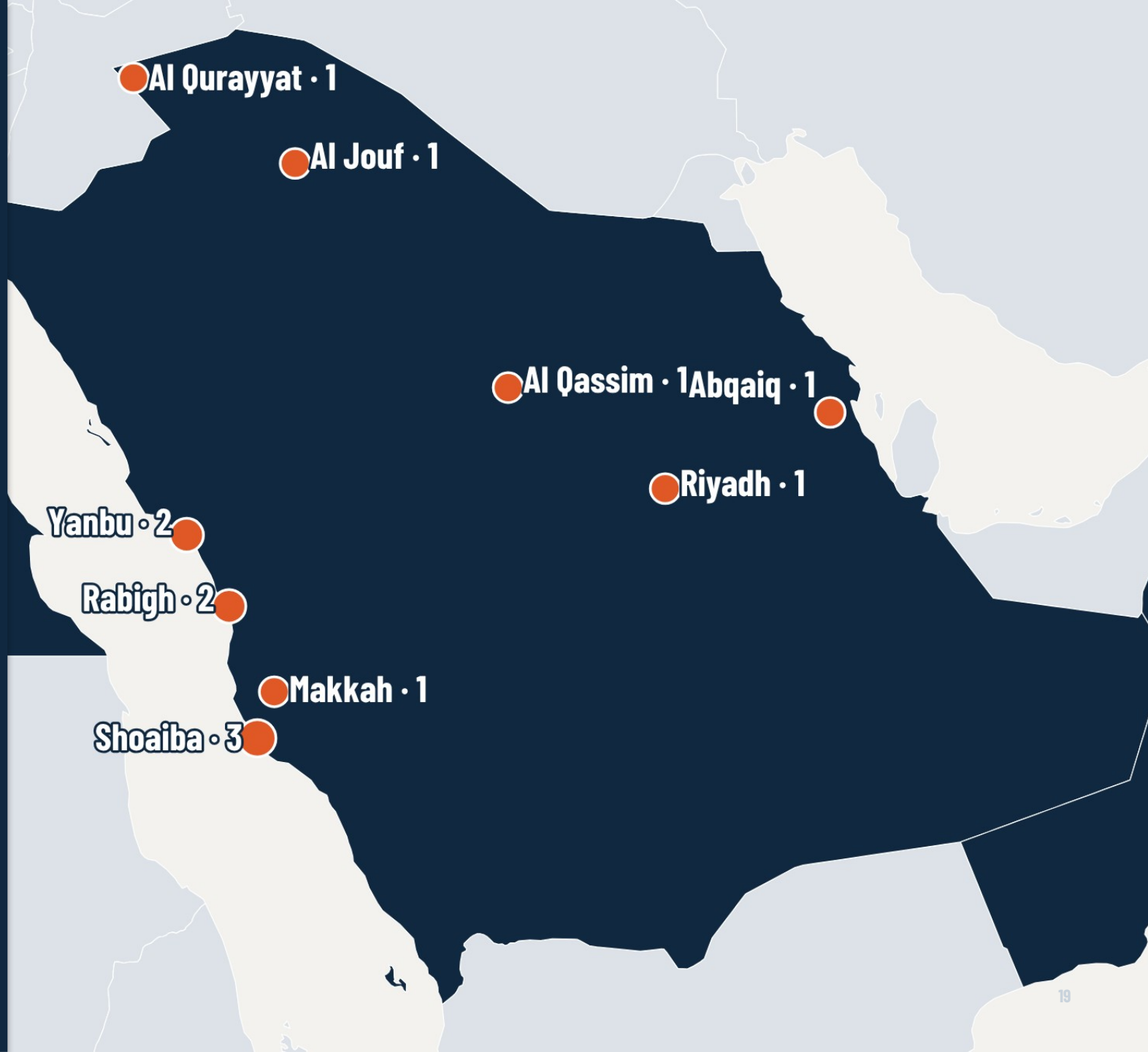
2009–2023

SAUDI ARABIA

13 REFERENCE PROJECTS

Shoaiba · Rabigh · Yanbu · Makkah · Riyadh · Al Qassim · Al Jouf · Al Qurayyat · Abqaiq

Steam and gas-turbine power plants, utilities, desalination and oil & gas processing.



SAUDI ARABIA · 2009–2010 · EMERSON DCS

SHOAIBA ST POWER PLANT 3×250 MW

DCS retrofit

CONTRACTOR

Emerson

END USER

Saudi Electricity Company

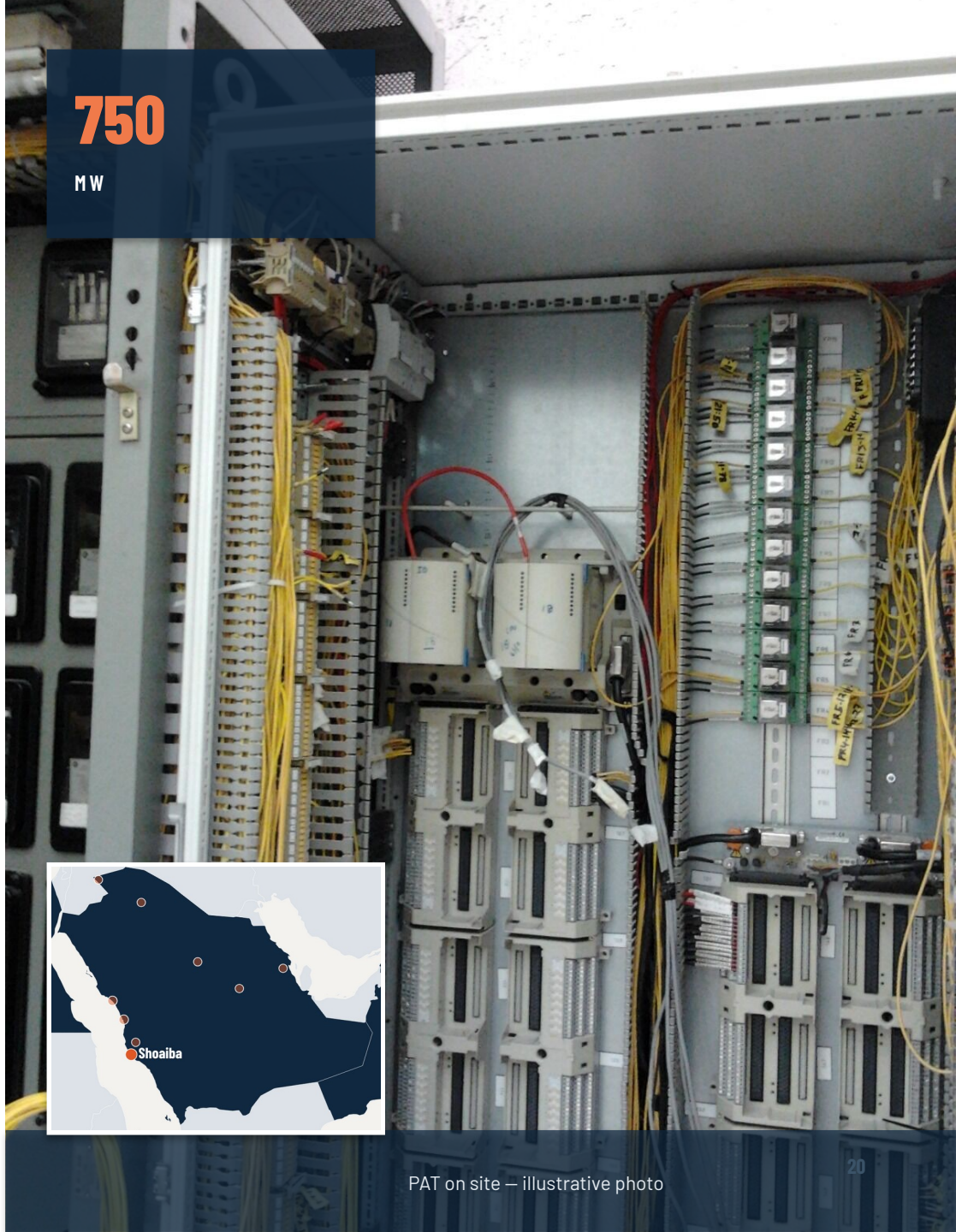
THE PROJECT

DCS retrofit of a steam power plant: IO loop and device checks, commissioning and start-up, returning the units to commercial operation.

PAT SCOPE

- Plant IO loop and device checks
- Commissioning and start-up, normalizing the system and returning to commercial operation

RESULT System normalised and units returned to commercial operation.



PAT on site — illustrative photo

SAUDI ARABIA · 2009 · EMERSON OVATION

PP4 RIYADH POWER PLANT — GE GAS TURBINES 2×25 MW

DCS retrofit

CONTRACTOR

EMERSON
Emerson

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم
Saudi Electricity Company

THE PROJECT

GE gas turbine loop and device checks, commissioning and start-up, plus DCS Ovation operation training for owner staff.

PAT SCOPE

- GE gas turbine loop and device checks, commissioning and start-up
- DCS Ovation operation training for owner staff
- Owner staff training courses on DCS plant operation and maintenance

RESULT Gas turbines commissioned and started up; owner operators trained on Ovation.



PAT on site — illustrative photo

SAUDI ARABIA · 2009–2011 · EMERSON DCS

RABIGH POWER PLANT 4×300 MW

DCS upgrade

CONTRACTOR

Emerson

END USER

Saudi Electricity Company

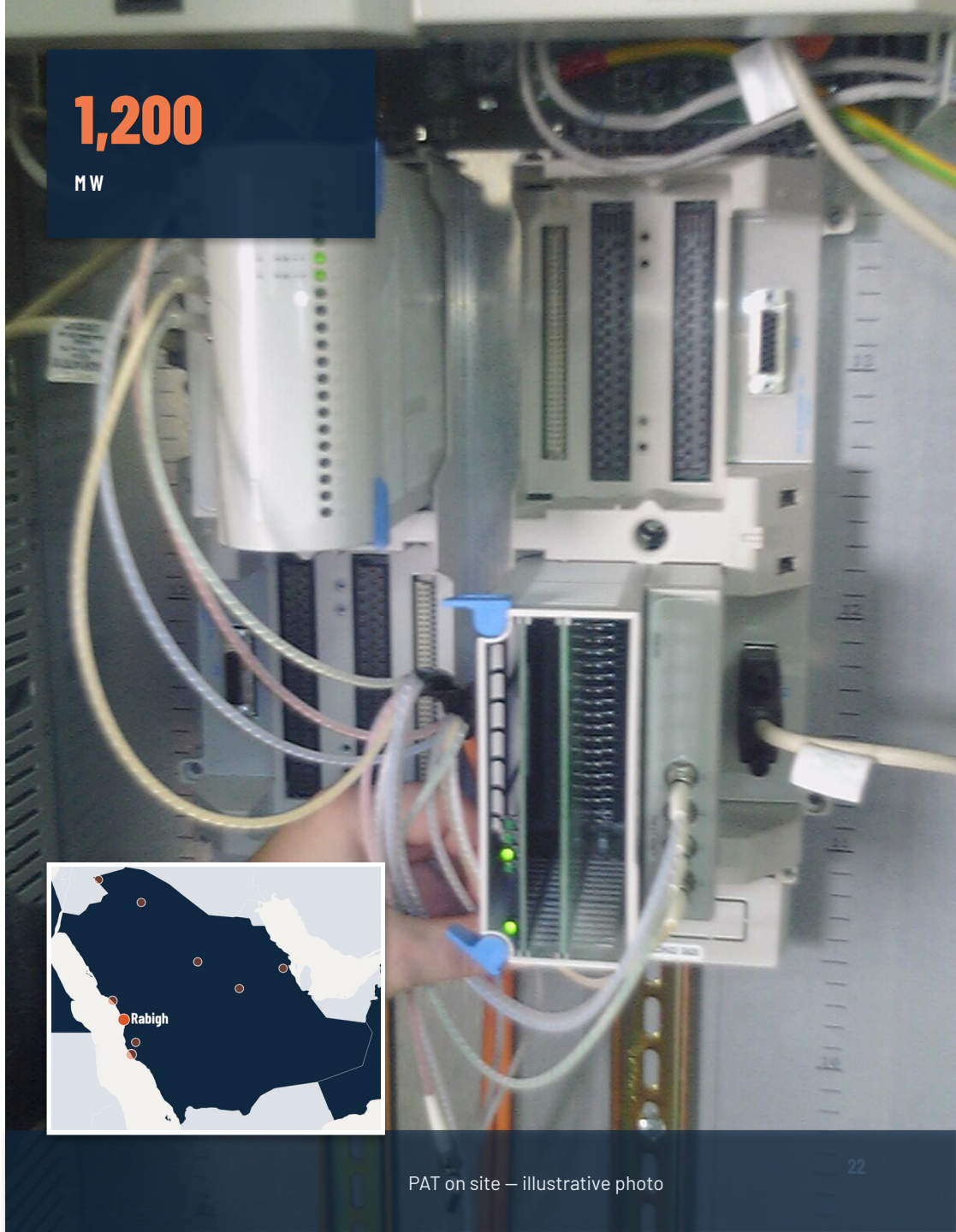
THE PROJECT

DCS software and hardware installation and programming, commissioning and start-up, ending with hand-over to the owner's staff.

PAT SCOPE

- DCS software and hardware installation and programming, commissioning, start-up and powering up
- Plant IO loop and device checks, commissioning and start-up, units normalized and handed over to the owner for commercial operation
- Owner staff training courses on DCS plant operation and maintenance

RESULT Units normalised and handed over to the owner for commercial operation.



PAT on site — illustrative photo

SAUDI ARABIA · 2009–2012 · EMERSON DCS

SEC4A AL JOUF — FIAT GAS TURBINES 7×35 MW

DCS upgrade

CONTRACTOR

EMERSON

Emerson

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم

Saudi Electricity Company

THE PROJECT

New DCS installation supervision and software installation, then gas turbine IO loop checks, commissioning and a guarantee-period scope.

PAT SCOPE

- Supervision of the new DCS installation and software installation
- Powering up the DCS and making it ready for gas turbine commissioning and start-up
- Gas turbine IO loop and device checks, plant commissioning and normalizing the units after the reliability test

RESULT Units back to commercial operation after the reliability test; guarantee period covered.



PAT on site — illustrative photo

SAUDI ARABIA · 2010 · EMERSON DCS

SHOAIBA LIME WATER PLANT

DCS engineering for FAT

CONTRACTOR

EMERSON
Emerson

END USER

Owner to be confirmed

THE PROJECT

Engineering of the database, logic and graphics software, ready for the factory acceptance test.

PAT SCOPE

- Engineering of database, logic and graphics software ready for FAT

RESULT Database, logic and graphics delivered ready for FAT.



PAT on site — illustrative photo

SAUDI ARABIA · 2013 · EMERSON DCS

SEC4A AL QURYAT — BBC GAS TURBINES 2×22 MW

DCS upgrade

CONTRACTOR

Emerson

END USER

Saudi Electricity Company

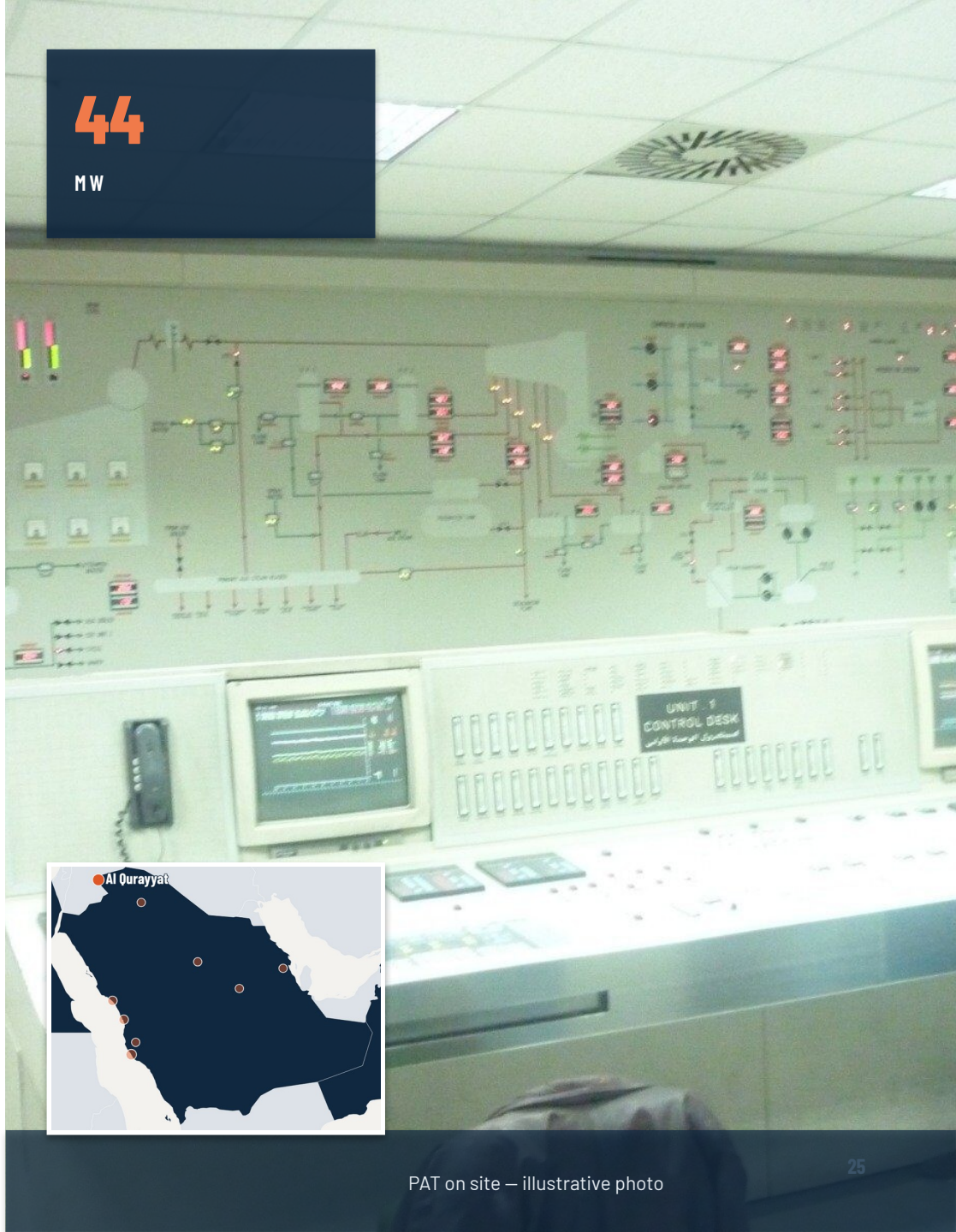
THE PROJECT

New DCS installation supervision and software installation, then gas turbine IO loop checks and commissioning.

PAT SCOPE

- Supervision of the new DCS installation and software installation
- Powering up the DCS and making it ready for gas turbine commissioning and start-up
- Gas turbine IO loop and device checks, plant commissioning and normalizing the units after the reliability test

RESULT Units back to commercial operation after the reliability test.



PAT on site — illustrative photo

SAUDI ARABIA · 2014 · EMERSON DCS

RABIGH II 2×300 MW

DCS modernization

CONTRACTOR

EMERSON
Emerson

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم
Saudi Electricity Company

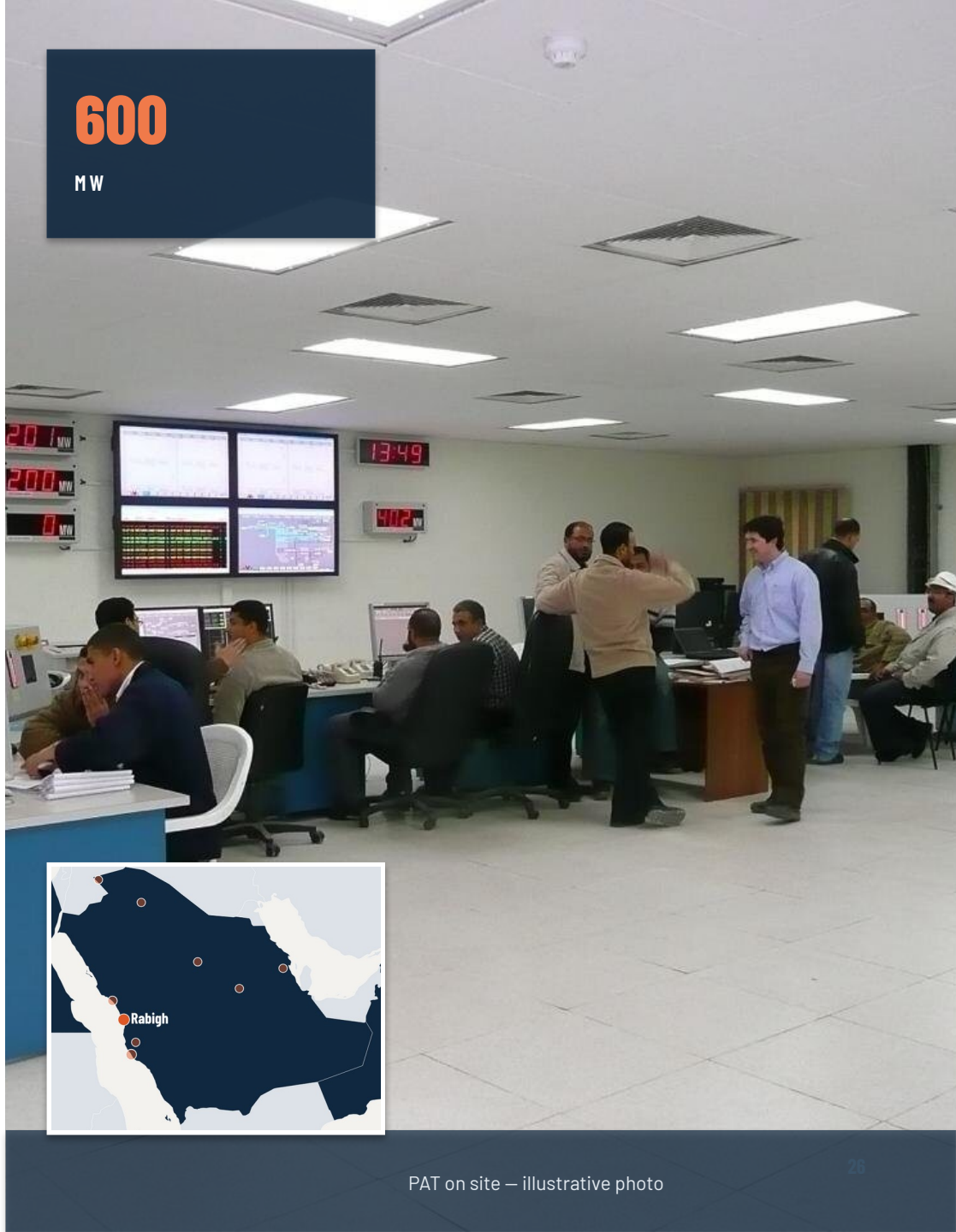
THE PROJECT

IO loop and device checks, commissioning and start-up after the DCS modernization, ending with hand-over to the owner.

PAT SCOPE

- Plant IO loop and device checks
- Commissioning and start-up, units normalized and returned to commercial operation by the owner (Saudi Electricity)

RESULT Units returned to commercial operation by the owner.



PAT on site — illustrative photo

SAUDI ARABIA · 2014–2017 · EMERSON DCS

MARAFIQ YANBU II 3×270 MW

New DCS

CONTRACTOR

EMERSON
Hanwha
Emerson · Hanwha

END USER

مرافق
MARAFIQ
Marafiq

THE PROJECT

DCS hardware and software modifications and system configuration, IO loop and device checks, commissioning and start-up, with warranty-period coverage by a PAT engineer.

PAT SCOPE

- DCS hardware and software modifications and system configuration
- Plant IO loop and device checks, commissioning and start-up
- Warranty period covered by a PAT engineer

RESULT New DCS commissioned; warranty period covered by a PAT engineer.



PAT on site — illustrative photo

SAUDI ARABIA · 2016 · EMERSON DCS

MAKKAH POWER PLANT — GE GAS TURBINES 6×65 MW

DCS upgrade — GT3 troubleshooting

CONTRACTOR

EMERSON
Emerson

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم
Saudi Electricity Company

THE PROJECT

GT3 start-up, operation troubleshooting and solution during the DCS upgrade of a GE gas turbine plant.

PAT SCOPE

- GT3 start-up, operation troubleshooting and solution

RESULT GT3 start-up and operation problems solved.



PAT on site — illustrative photo

SAUDI ARABIA · 2017

AL QASEEM POWER STATION — WESTINGHOUSE GAS TURBINES 9×50 MW

GT9 valve replacement & commissioning

CONTRACTOR

SAUDI FAL

Saudi FAL

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم

Saudi Electricity Company

THE PROJECT

Supervision of the new FCV installation (GT9 FCV, OST and HP/LP valves replaced with new REGZA valves), software installation and commissioning.

PAT SCOPE

- Supervision of the new FCV installation and software installation, ready for gas turbine commissioning and start-up
- GT9 IO loop and device checks, plant commissioning and normalizing the unit after the reliability test

RESULT GT9 back to commercial operation after the reliability test.



PAT on site — illustrative photo

SAUDI ARABIA · 2019

YANBU I & II — MARAFIQ TRAINING ACADEMY

Consulting & training

CONTRACTOR

L3ORON Institute

Leoron

END USER

مرافق MARAFIQ

Marafiq

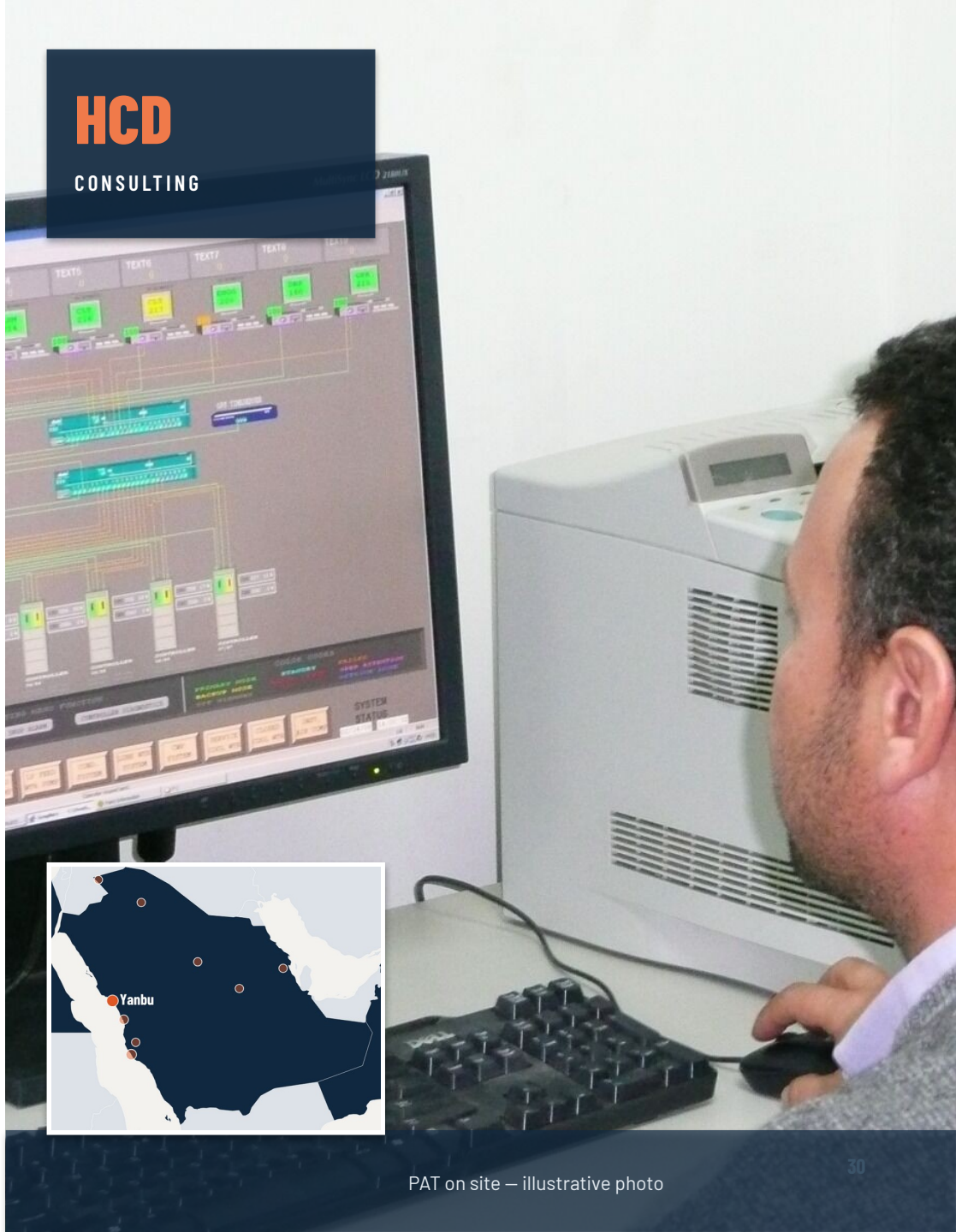
THE PROJECT

HCD senior consultant for Marafiq's utility and engineering practice: improved work-flow procedures and established a training academy.

PAT SCOPE

- Work-flow procedure improvements
- Established an academy for the Marafiq training center
- Created training course material for engineers, supervisors and technicians on operation and maintenance

RESULT Training academy established for Marafiq's training centre.



PAT on site — illustrative photo

SAUDI ARABIA · 2021-2022 · EMERSON OVATION

SHOAIBA POWER PLANT, STAGE I PHASE 1

DCS rehabilitation — ABB to Emerson Ovation

CONTRACTOR

EMERSON
Emerson · Giza Systems

GIZA SYSTEMS

END USER

الشركة السعودية للكهرباء
Saudi Electricity Company
نعمل بإتقان من أجلكم

Saudi Electricity Company

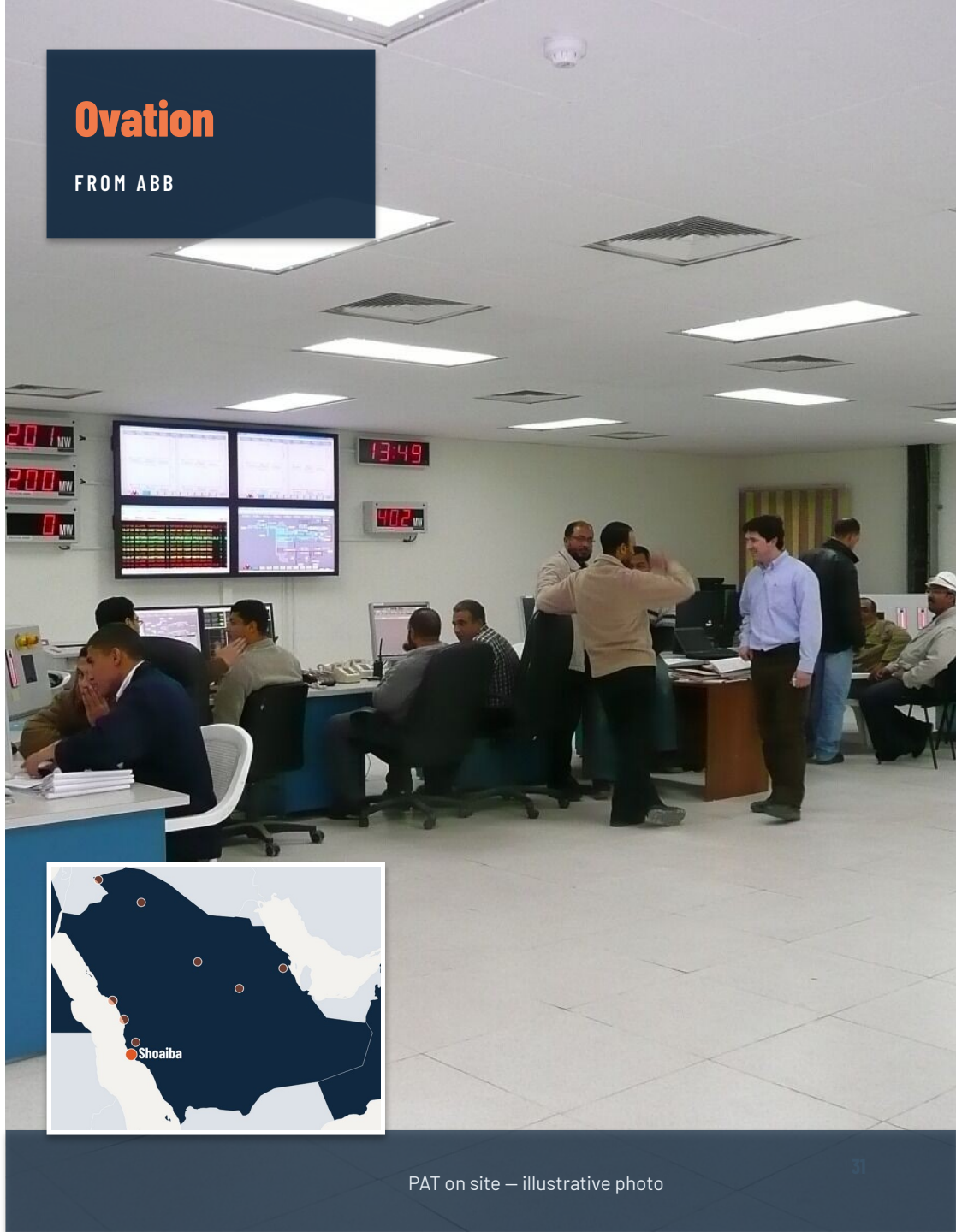
THE PROJECT

DCS rehabilitation from ABB to Emerson Ovation on a very tight schedule: IO loop checks, device tests and commissioning of boiler, turbine, BOP, desalination and demi water.

PAT SCOPE

- IO loop checks, device tests and commissioning of the boiler, turbine, BOP, desalination and demi water systems
- Troubleshooting of the Ovation DCS together with the Emerson team
- Comprehensive support throughout the project on a tight construction schedule to return the plant to commercial operation

RESULT Delivered on a very tight schedule to return the plant to commercial operation.



PAT on site — illustrative photo

SAUDI ARABIA · 2022–2023 · EMERSON DELTAV

ABQAIQ PLANTS — DCS UPGRADE, PIB-2 TO PIB-18 (ARAMCO)

DCS upgrade — DeltaV

CONTRACTOR

EMERSON · SAUDI FAL · KBR

Emerson · Saudi FAL · KBR (design)

END USER

أرامكو السعودية
saudi aramco

Saudi Aramco

THE PROJECT

DCS upgrade across twelve PIBs: document review, cabinet staging checks before FAT, construction supervision and hot and cold tests for each PIB shutdown.

PAT SCOPE

- Engineering document review and verification of cabinet and marshalling staging before FAT
- Supervision of construction activities
- Hot and cold test commissioning for each PIB shutdown

RESULT Hot and cold tests completed for each of the twelve PIB shutdowns.



PAT on site — illustrative photo

2014

OMAN

1 REFERENCE PROJECT

Sur

A 2,000 MW combined-cycle IPP — Ovation DCS and Automatic Plant Start.



OMAN · 2014 · EMERSON OVATION

SUR IPP-CCPP 2,000 MW

Ovation DCS — APS check & commissioning

CONTRACTOR

EMERSON

DAEWOO E&C

Emerson · Daewoo E&C

END USER

PHOENIX POWER

Phoenix Power Company

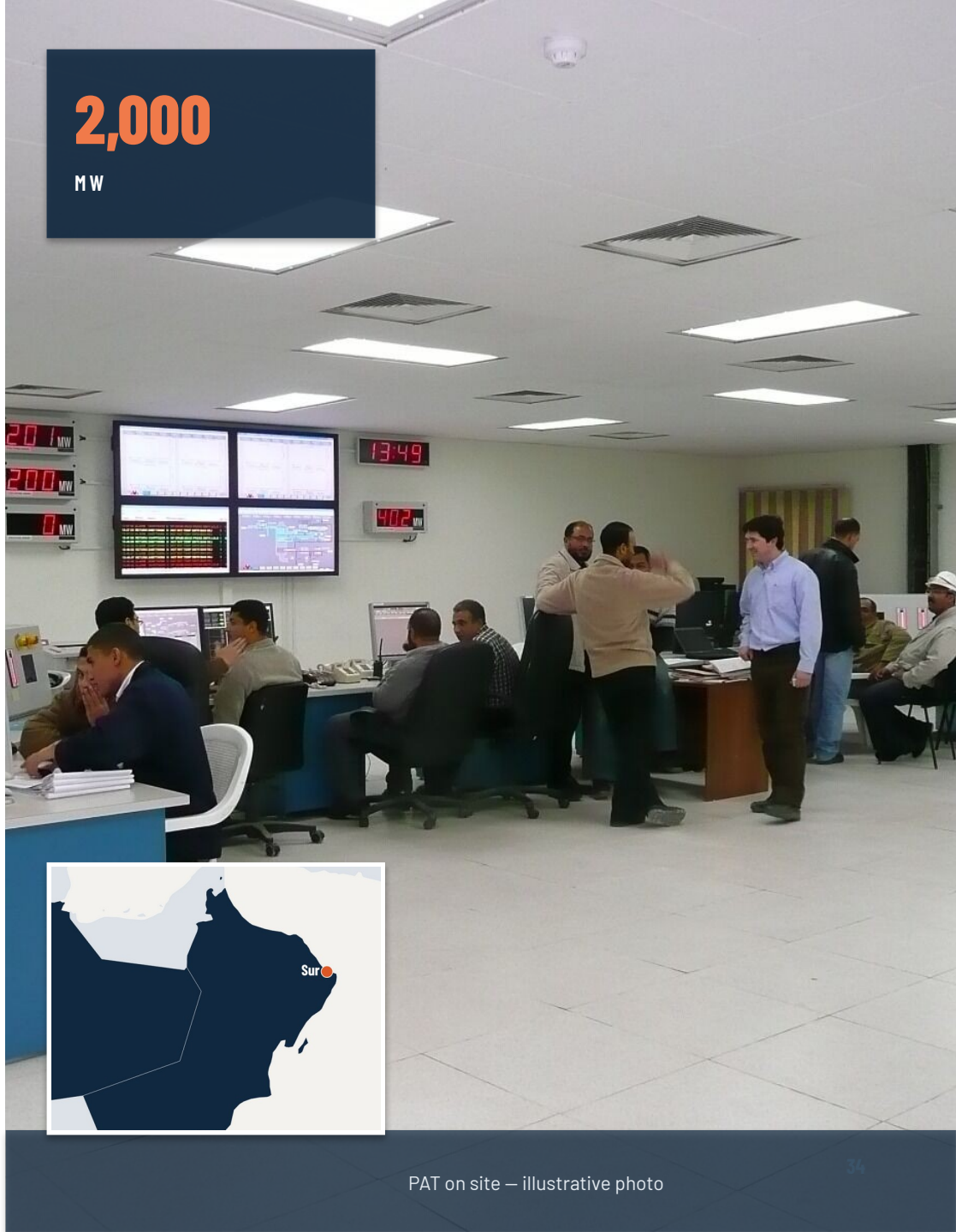
THE PROJECT

Check and modification of the Automatic Plant Start (APS) software on the new Ovation DCS so the plant starts as one power block.

PAT SCOPE

- Check, troubleshooting and modification of the APS (Automatic Plant Start) software so it works as one power block plant start
- Plant systems commissioning and start-up

RESULT Automatic Plant Start working as one power-block start; plant systems commissioned.



PAT on site — illustrative photo



RECOGNITION

RECOGNISED BY EMERSON PROCESS MANAGEMENT

"In recognition of Ahdy Abu-Malwa for technical contributions and excellence in delivery of Emerson Ovation Expert Control System Solutions within the Arab Republic of Egypt."

DCS LEAD ENGINEER

New Talkha 750 MW CC Expansion

Damietta 1,230 MW WDPF to Ovation Migration

El-Kureimat II 750 MW CC Power Station

DCS ENGINEER

Ataka 900 MW Steam Modernization

Signed by Robert Yeager, President, Emerson PWS, and T.C. Chan, Project Manager, Emerson PWS. Emerson and Ovation are trademarks of Emerson Electric Co.

POWER AUTOMATION TEAM

DCS & INSTRUMENTS SPECIALISTS

YOUR NEXT OUTAGE OR START-UP

LET'S BRING YOUR CONTROL SYSTEM ONLINE

PHONE +20 109 991 2020 (24/7)

E-MAIL info@patdcs.com · sales@patdcs.com

WEB patdcs.com

OFFICE New Damietta, Egypt

Scan for the full project list